

Labsheet-9

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
df = pd.read_csv(next(iter(uploaded)))
```

```
print(df.shape)
```

```
df.head()
```

Choose Files country_wis...olumns.csv

country_wise_latest-selected-columns.csv(text/csv) - 8918 bytes, last modified: 9/16/2026 - 100% done Saving country_wise_latest-selected-columns.csv to country_wise_latest-selected-columns (1).csv (187, 10)

Country/Region	Recovered	Active	New deaths	recovered Deaths / 100 Cases	Recovered / 100 Cases																								
Confirmed Deaths	New cases	New																											
0 Afghanistan	36263	1269	25198	9796	106	10	18	3.50	69.49	1 Albania	4880	144	2745	1991	117	6	63	2.95	56.25										
2 Algeria	27973	1163	18837	7973	616	8	749	4.16	67.34	3 Andorra	907	52	803	52	10	0	0	5.73	88.53	4 Angola	950	41	242	667	18	1	0	4.32	25.47

Next steps: [Generate code with df](#)
[New interactive sheet](#)

```
df.columns = df.columns.str.strip()
print(df.columns.tolist())
df.info()
```

```
['Country/Region', 'Confirmed', 'Deaths', 'Recovered', 'Active', 'New cases', 'New deaths', 'New recovered', 'Deaths / 100 C <class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 187 entries, 0 to 186
```

```
Data columns (total 10 columns):
```

```
# Column Non-Null Count Dtype
```

```
--- ---
```

```
0 Country/Region 187 non-null object
```

```
1 Confirmed 187 non-null int64
```

```
2 Deaths 187 non-null int64
```

```
3 Recovered 187 non-null int64
```

```
4 Active 187 non-null int64
```

```
5 New cases 187 non-null int64
```

```
6 New deaths 187 non-null int64
```

```
7 New recovered 187 non-null int64
```

```
8 Deaths / 100 Cases 187 non-null float64
```

```
9 Recovered / 100 Cases 187 non-null float64
```

```
dtypes: float64(2), int64(7), object(1)
```

```
memory usage: 14.7+ KB
```

```
print("Missing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
```

Missing Values:

Country/Region 0

Confirmed 0

Deaths 0

Recovered 0

Active 0

New cases 0

New deaths 0

New recovered 0

Deaths / 100 Cases 0

Recovered / 100 Cases 0

```
dtype: int64
```

```
Duplicate Rows: 0
```

```
date_cols = [c for c in df.columns if "date" in c.lower()]
print("Date columns:", date_cols)
```

```
if date_cols:
    df[date_cols[0]] = pd.to_datetime(df[date_cols[0]], errors="coerce")
```

```
Date columns: []
```

```
num_cols = df.select_dtypes(include=np.number).columns.tolist()
print("Numerical Columns:", num_cols)
print(df[num_cols].describe())
```

```
Numerical Columns: ['Confirmed', 'Deaths', 'Recovered', 'Active', 'New cases', 'New deaths', 'New
recovered', 'Deaths / 100', 'Confirmed Deaths Recovered Active New cases \
count 1.870000e+02 187.000000 1.870000e+02 1.870000e+02 187.000000
mean 8.813094e+04 3497.518717 5.063148e+04 3.400194e+04 1222.957219
std 3.833187e+05 14100.002482 1.901882e+05 2.133262e+05 5710.374790
min 1.000000e+01 0.000000 0.000000e+00 0.000000e+00 0.000000
25% 1.114000e+03 18.500000 6.265000e+02 1.415000e+02 4.000000
50% 5.059000e+03 108.000000 2.815000e+03 1.600000e+03 49.000000
75% 4.046050e+04 734.000000 2.260600e+04 9.149000e+03 419.500000
max 4.290259e+06 148011.000000 1.846641e+06 2.816444e+06 56336.000000
```

```
    New deaths New recovered Deaths / 100 Cases Recovered / 100 Cases
count 187.000000 187.000000 187.000000 187.000000
mean 28.957219 933.812834 3.019519 64.820535
std 120.037173 4197.719635 3.454302 26.287694
min 0.000000 0.000000 0.000000 0.000000
25% 0.000000 0.000000 0.945000 48.770000
50% 1.000000 22.000000 2.150000 71.320000
75% 6.000000 221.000000 3.875000 86.885000
max 1076.000000 33728.000000 28.560000 100.000000
```

```
if date_cols and num_cols:
    date_col = date_cols[0]
    value_col = num_cols[0]
    trend = df.groupby(date_col)[value_col].sum()
```

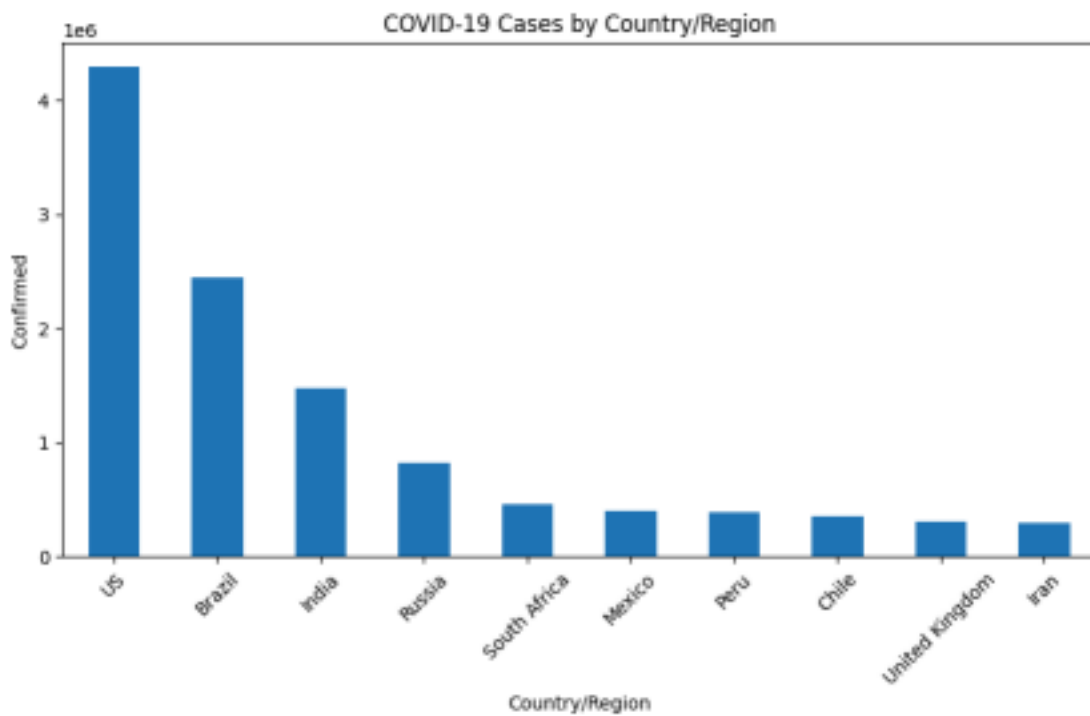
```
plt.figure(figsize=(12,5))
plt.plot(trend)
plt.title("COVID-19 Trend Over Time")
plt.xlabel("Date")
plt.ylabel(value_col)
plt.xticks(rotation=45)
plt.show()
```

```
cat_cols = df.select_dtypes(include="object").columns.tolist()
print("Categorical Columns:", cat_cols)
```

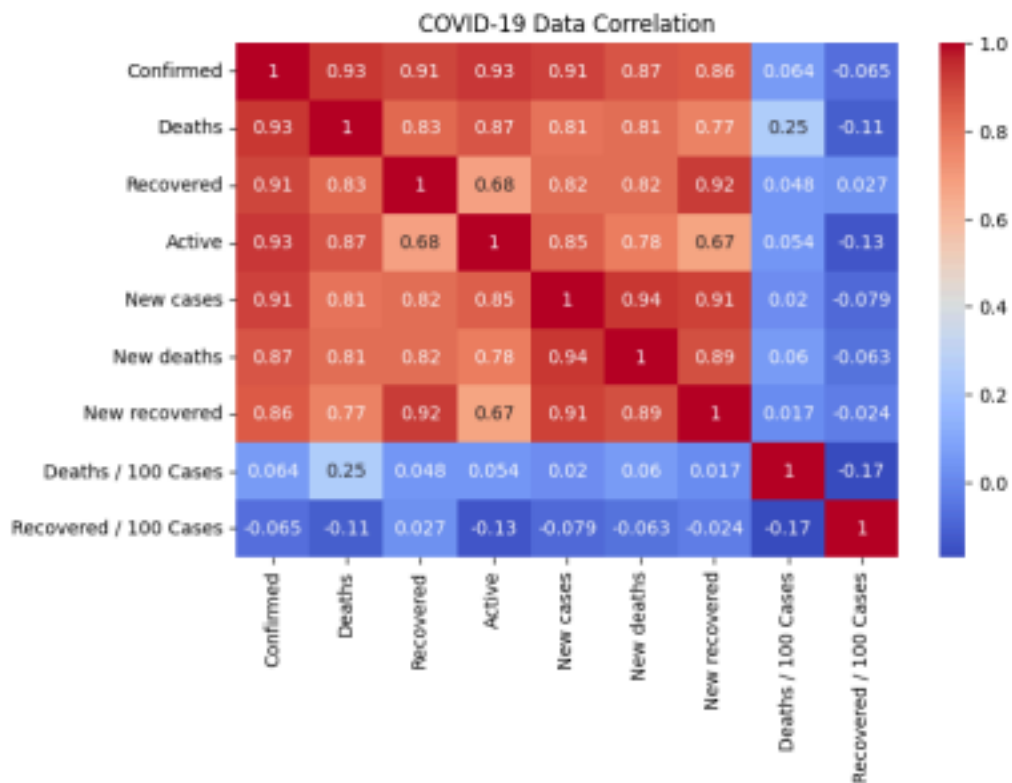
```
Categorical Columns: ['Country/Region']
```

```
if cat_cols and num_cols:
    col = cat_cols[0]
    value_col = num_cols[0]
    data = df.groupby(col)[value_col].sum().sort_values(ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))
data.plot(kind="bar")
plt.title("COVID-19 Cases by " + col)
plt.xlabel(col)
plt.ylabel(value_col)
plt.xticks(rotation=45)
plt.show()
```



```
if len(num_cols) >= 2:
    plt.figure(figsize=(8,5))
    sns.heatmap(df[num_cols].corr(), annot=True, cmap="coolwarm")
    plt.title("COVID-19 Data Correlation")
    plt.show()
```



```
if num_cols:
    totals = df[num_cols].sum().sort_values(ascending=False)
    print("COVID-19 KPIs")
    print(totals)
```

```
COVID-19 KPIs
Confirmed 16480485.00
Recovered 9468087.00
Active 6358362.00
Deaths 654036.00
New cases 228693.00
New recovered 174623.00
Recovered / 100 Cases 12121.44
```

New deaths 5415.00
Deaths / 100 Cases 564.65
dtype: float64

```
print("BUSINESS INSIGHTS")  
print("1. COVID-19 trends can be monitored over time.")  
print("2. High-case regions can be identified for resource allocation.")  
print("3. Correlation between numerical variables can reveal important patterns.")  
print("4. COVID-19 dashboards can support healthcare and operational planning.")
```

BUSINESS INSIGHTS

1. COVID-19 trends can be monitored over time.

Theory Questions & Answers

1. What is data storytelling? How is it different from data visualization?

Data storytelling is the process of combining data, visualizations, and a narrative to communicate meaningful insights. It explains not only **what happened**, but also **why it matters** and **what action can be taken**.

Data visualization mainly represents data in graphical form such as charts, graphs, and dashboards.

Thus, visualization presents the **information**, while storytelling provides the **context, meaning, and conclusion**.

2. Why is storytelling important in business analytics and decision-making?

Storytelling makes complex analytical results easier to understand for business stakeholders. Instead of presenting only numbers and charts, it explains the important trends, problems, and opportunities.

It helps organizations to:

- Communicate insights clearly.
- Identify important business problems.
- Support data-driven decisions.
- Make presentations more engaging.
- Convert analytical findings into actionable strategies.

3. What are the essential components of an effective data story?

An effective data story generally contains:

1. **Data** – Reliable and relevant information.
2. **Context** – Background and business problem.
3. **Visualizations** – Charts or graphs that communicate important findings.
4. **Narrative** – Logical explanation of the findings.
5. **Insights** – Meaning derived from the data.
6. **Recommendations** – Possible actions based on the findings.

These components together create a clear and meaningful data-driven story.

4. How do Key Performance Indicators (KPIs) enhance business reporting?

KPIs are measurable values used to evaluate the performance of a business, department, or process.

Examples include:

- Total Sales
- Profit
- Revenue Growth
- Customer Retention Rate
- Conversion Rate

KPIs make reports more focused because stakeholders can quickly understand whether important business objectives are being achieved. They also help in comparing performance across different periods or

departments.

5. Why should visualizations be arranged in a logical sequence?

Visualizations should follow a logical sequence so that the audience can easily understand the complete analytical story.

A common sequence is:

Problem → Data → Trend → Analysis → Insight → Recommendation

For example, a sales dashboard may first show total sales, then sales trends, followed by regional performance and finally the areas requiring attention.

A logical sequence prevents confusion and helps stakeholders connect different findings.

6. What factors should be considered while selecting visualizations for a business presentation?

The visualization should be selected according to the type of information being communicated.

Important factors include:

- **Type of data:** categorical, numerical, time-series, etc.
- **Purpose:** comparison, trend, distribution, or relationship.
- **Audience:** technical or non-technical.
- **Clarity:** visualization should be easy to understand.
- **Amount of data:** avoid overcrowded charts.
- **Accuracy:** visualization should represent data correctly.
- **Business relevance:** it should highlight information useful for decision-making.

For example, a line chart is suitable for showing sales trends over time, while a bar chart is useful for comparing sales between categories.

7. Explain how dashboards and storytelling complement each other in Business Intelligence.

A **dashboard** presents important business information through interactive charts, graphs, KPIs, and filters. **Storytelling** provides the narrative that explains the meaning of these visualizations.

For example:

Dashboard: Shows that sales decreased in a particular region.

Storytelling: Explains when the decrease occurred, possible patterns associated with it, and what business areas may require further investigation.

Therefore, dashboards provide the **interactive evidence**, while storytelling provides the **context and interpretation**.

8. What challenges may arise while communicating analytical insights to

non-technical stakeholders?

Common challenges include:

- Lack of understanding of technical terminology.
- Too many charts or excessive information.
- Difficulty understanding statistical concepts.
- Presenting insights without sufficient business context.
- Different stakeholders having different priorities.
- Misinterpretation of charts or metrics.

These challenges can be reduced by using simple language, clear visualizations, relevant KPIs, and business-focused explanations.

9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Example 1 – Retail Business:

A retail company can analyze sales data and present a story showing that certain products have consistently higher demand during particular periods. Management can use this insight for inventory planning and promotional campaigns.

Example 2 – Public Health:

During the COVID-19 pandemic, dashboards and visual data stories were used to communicate case trends, hospital demand, vaccination coverage, and other indicators. Such presentations helped authorities and the public understand changing conditions and evaluate response measures.

10. How can effective data storytelling improve strategic planning and organizational performance?

Effective data storytelling converts complex analytical results into information that decision-makers can understand and act upon.

It can help organizations:

- Identify trends and performance gaps.
- Understand business opportunities and risks.
- Set measurable objectives.
- Allocate resources more effectively.
- Monitor KPIs and progress.
- Make evidence-based strategic decisions.
- Communicate goals clearly across departments.

Therefore, effective data storytelling creates a connection between **data analysis and business action**, helping organizations use analytical findings more effectively.